EXPERIMENT – 4.1

Consider the following table: Table: CLASS

ID Name

1 Bravo

2 Alex

4 Cheng

Give the output of the following SQL script:

• INSERT INTO class VALUES (5,'Rahul');

• COMMIT;

• UPDATE class SET name = 'Abhijeet' WHERE id= '5';

• SAVEPOINT A;

• INSERT INTO class VALUES (6, 'Chris');

• SAVEPOINT B;

• INSERT INTO class VALUES (7, 'Bravo');

• SAVEPOINT C

• SELECT * FROM class;

• ROLLBACK TO B;

• SELECT * FROM class;

• ROLLBACK TO A;

CODE:

```sql
CREATE DATABASE Assignment4DB;
USE Assignment4DB;

CREATE TABLE class (
    id INT PRIMARY KEY,
    name VARCHAR(50)
);

INSERT INTO class VALUES (1,'Bravo');
INSERT INTO class VALUES (2,'Alex');
INSERT INTO class VALUES (4,'Cheng');
COMMIT;

INSERT INTO class VALUES (5,'Rahul');
COMMIT;

UPDATE class SET name = 'Abhijeet' WHERE id = 5;
SAVEPOINT A;

INSERT INTO class VALUES (6,'Chris');
```

```sql
SAVEPOINT B;

INSERT INTO class VALUES (7,'Bravo');
SAVEPOINT C;

SELECT * FROM class;

ROLLBACK TO B;

SELECT * FROM class;

ROLLBACK TO A;

SELECT * FROM class;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

class 1 | Apply | Revert

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

class 2 | Apply | Revert

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

class 3 | Apply | Revert

EXPERIMENT -4.2

Consider the following table: Table: ACCOUNT_TXN

Acc_ID Holder_Name Balance

101 Aman 5000

102 Bhavna 7000

103 Chirag 6000

Perform the following TCL-based SQL script and write the output after each SELECT statement:

```
INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

SAVEPOINT S1;

UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101;

UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102;

SAVEPOINT S2;

DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103;

INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);

SAVEPOINT S3;

UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S2;

SELECT * FROM ACCOUNT_TXN;

INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500);

SAVEPOINT S4;

UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN;
```

```
ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

COMMIT;
```

Student Tasks

Write the output of all 4 SELECT statements in tabular format.

Identify which rows are permanently saved after COMMIT.

Mention which operations were undone by:

ROLLBACK TO S2

ROLLBACK TO S4

Write the final state of the table after COMMIT.

Additional Challenge (Conceptual)

Answer the following:

What is the difference between ROLLBACK and ROLLBACK TO SAVEPOINT?

Can we rollback to S3 after ROLLBACK TO S2? Why/Why not?

What happens if COMMIT is executed before creating any savepoint?

CODE:

```sql
CREATE DATABASE Assignment42DB;
USE Assignment42DB;

CREATE TABLE ACCOUNT_TXN (
    Acc_ID INT PRIMARY KEY,
    Holder_Name VARCHAR(50),
    Balance INT
);

INSERT INTO ACCOUNT_TXN VALUES (101,'Aman',5000);
INSERT INTO ACCOUNT_TXN VALUES (102,'Bhavna',7000);
INSERT INTO ACCOUNT_TXN VALUES (103,'Chirag',6000);

INSERT INTO ACCOUNT_TXN VALUES (104,'Diya',4000);

SAVEPOINT S1;

UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101;
UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102;
```

```
SAVEPOINT S2;

DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103;
INSERT INTO ACCOUNT_TXN VALUES (105,'Eshan',3000);

SAVEPOINT S3;

UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S2;

SELECT * FROM ACCOUNT_TXN;

INSERT INTO ACCOUNT_TXN VALUES (106,'Farah',4500);

SAVEPOINT S4;

UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

COMMIT;
```

OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna Sharma	6000
104	Diya	4000
105	Eshan	3000

COUNT_TXN 1 | Apply | Revert

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna Sharma	6000
104	Diya	4000
105	Eshan	3000
NULL	NULL	NULL

ACCOUNT_TXN 2 × Apply Revert

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	7000
102	Bhavna Sharma	6000
104	Diya	4500
105	Eshan	3000
106	Farah	4500
NULL	NULL	NULL

ACCOUNT_TXN 3 × Apply Revert

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	7000
102	Bhavna Sharma	6000
104	Diya	4500
105	Eshan	3000
106	Farah	4500
NULL	NULL	NULL

ACCOUNT_TXN 4 × Apply Revert